DSA5205 Project 2

Cross Sector Information and Sector ETF Trading

Linear and nonlinear return forecasts with cost aware portfolio evaluation

Group 12

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Abstract

We examine whether cross-sector information improves daily forecasts and trading performance for nine US sector ETFs relative to SPY. S0 contains own-sector and common-market predictors; S1 adds other sectors' return histories; S2 adds relationship summaries. Elastic Net and GBDT use monthly expanding windows, with 2018–2022 validation and 2023–September 2026 testing. Forecasts target next-open-to-following-open sector-minus-SPY cash returns, with training restricted to labels available before each monthly model refit.

Elastic Net produces identical test forecasts across specifications, with OOS R^2 of -0.047%; strong shrinkage and XLF dominate its weak performance. GBDT S1 achieves 0.043%, but test loss-improvement intervals include zero. S2 does not consistently improve on S1.

After costs, the GBDT S1 cash-based pair strategy earns 5.534% annualized versus 4.808% for cash. The SPY-core overlay earns 22.723% versus 22.019% for dividend-reinvestment SPY. Its high absolute return reflects continuous market exposure. The overlay was introduced after test results were examined, and its reported configuration was selected by test performance; these findings are exploratory. Overall, modest forecast differences translate into benchmark-dependent investment outcomes.

1 Introduction

Sector returns reflect connected cash flows. Higher energy prices raise costs for transport and manufacturing, while tighter financing conditions influence investment, borrowing and household demand across industries. Investors must interpret these shocks and assess their effects on different firms. Limited attention and gradual information processing can cause related sectors to incorporate the same economic news at different speeds, creating a possible role for other sectors' recent returns in forecasting a target sector (Rapach et al., 2019).

Rapach et al. (2019) study thirty industry portfolios using monthly returns and find predictive information in other industries' lagged returns. Chinco et al. (2019) identify sparse, short-lived signals for one-minute-ahead stock returns using a large set of lagged returns. We examine an intermediate horizon: daily forecasts for nine tradable US sector ETFs, with the target defined as the next-open-to-following-open cash return relative to SPY. This setting connects cross-industry predictability to instruments that can form sector–market trading pairs.

Building on this evidence, we compare three nested feature specifications and two model families to test the following hypotheses:

H1 Cross-sector information: adding the other eight sectors' return histories in S1 improves out-of-sample prediction relative to S0, which contains own-sector information and common market controls.

H2 Relationship representations: adding sector rankings, dispersion and correlation summaries in S2 improves prediction relative to S1.

H3 Nonlinearity: gradient boosted decision trees improve prediction relative to Elastic Net on the same specification and evaluation sample. Nonlinear models allow the effect of a sector signal to vary with market conditions, such as volatility and cross-sector co-movement. Predictive gains are assessed by lower mean squared error on common forecast-scoring samples.

Section 2 defines the target, information timing and comparison protocol. Section 3 describes the assets and source data. Section 4 presents cleaning, return verification, feature construction and the resulting panels. Subsequent sections report model estimation and forecast results, followed by portfolio implementation and evaluation.

2 Research Design

2.1 Target and information timing

For adjacent opening dates a and b, the return including cash dividends is

$$R^{\text{cash}}_{i,a\to b} = \frac{O_{i,b} + D_{i,b}}{O_{i,a}} - 1 \tag{1}$$

O is the provider's split-normalized opening price without dividend adjustment, and D is the cash dividend per

share in matching units. Dividends accrue on the ex-dividend date, so a position held before the exit-date open receives that date's dividend. The prediction target for sector i on signal date t is

$$y_{i,t} = R^{\text{cash}}_{i,t+1\to t+2} - R^{\text{cash}}_{\text{SPY},t+1\to t+2} \tag{2}$$

Signals use observations through t's close and are formed at 18:00 New York time. Entry and settlement occur at the t+1 and t+2 opens; offsets are SPY sessions and returns are decimals. Positive targets indicate sector outperformance. The equal-dollar sector–SPY difference retains beta exposure and constituent overlap because SPY includes the target sector.

2.2 Nested specifications and sector models

S0 combines own-sector information with common controls. S1 adds eight individually named sectors' return histories. S2 adds relationship summaries: ranks and dispersion re-express S1 information, while correlations use sixty-day paths. Separate models allow predictive links to differ by target sector. Elastic Net regularizes correlated predictors (Zou and Hastie, 2005); boosted trees capture nonlinear responses and interactions. Both families require complexity control given one daily observation per sector.

2.3 Chronological protocol and monthly refitting

Table 1: Sample periods and protocol roles

Period	Protocol role
2006–2007	Feature warm-up
2008–2017	Initial estimation history
2018–2022	Forward validation and parameter selection
2023–30 September 2026	Out-of-sample forecast evaluation

Models are refitted at 18:00 New York time on the last SPY session before each forecast month, expanding from 2008. Eligible observations have complete predictors and valid targets, with signal and label availability no later than the fitting cutoff. A signal dated s becomes eligible at the s+2 open, when its full holding-period return is observed. This label-maturity purge excludes the two latest signal dates: the first fit on 29 December 2017 uses signals through 27 December. Model and fitted preprocessing parameters remain fixed within each forecast month.

2.4 Common comparison samples

Both model families and all three specifications use the same eligible observations within each sector. Scoring additionally requires the holding interval to end within its designated split. This holds date coverage constant; the resulting loss of otherwise usable S0 observations is quantified in Section 4.3.

3 Data Sources

3.1 Asset universe and roles

The nine established sector funds in Table 2 provide a long common history with interpretable exposures. XLRE and XLC began on 7 October 2015 and 18 June 2018, respectively; their shorter histories motivate exclusion.[1]

3.2 Acquisition and frozen snapshots

The requested daily range is 1 January 2006–30 September 2026. Yahoo Finance, through yfinance, supplies OHLC, volume, adjusted close, dividends and split records for fifteen ETFs. Cboe supplies VIX's historical CSV; FRED's observation table supplies DGS3MO, an annualized percent yield.[2]

The submission includes forty-seven unchanged snapshots: raw and adjusted CSVs plus Yahoo JSON for each ETF, and the VIX and DGS3MO files. All SHA-256 hashes match the download manifest, as verified in

[1] State Street, XLRE and XLC fund profiles; accessed 4 November 2026.

[2] Yahoo Finance; Cboe VIX history; FRED DGS3MO. Snapshots retrieved 20 October 2026.

Table 2: Asset universe and roles

Role	Assets	Purpose
Target sector ETFs	XLB Materials; XLE Energy; XLF Financials; XLI Industrials; XLK Technology; XLP Consumer Staples; XLU Utilities; XLV Health Care; XLY Consumer Discretionary	Prediction targets and tradable legs paired with SPY
Market ETF	SPY	Master calendar, relative-return reference, investable benchmark and paired trading leg
Auxiliary ETFs	TLT, IEF, GLD, HYG, LQD	Treasury, gold and corporate-bond market conditions
Market volatility	Cboe VIX	Common volatility conditions
Interest rate	FRED DGS3MO	Cash and financing calculations

01_download_clean_data.ipynb. Snapshot reuse fixes the inputs; network reacquisition may reflect provider revisions. Adjusted versions serve as diagnostic references; modelling uses dividend-unadjusted prices and explicit distributions.

3.3 Sample coverage

SPY provides 5,218 sessions from 3 January 2006. The common scoring sample has 41,657 date–sector observations: 21,911 in initial estimation, 11,313 in validation and 8,433 in testing. Initial sector counts are 2,456 for eight ETFs and 2,263 for XLF. Sector and monthly sample eligibility are reported in 02_features_panel_data.ipynb.

4 Data Cleaning and Feature Construction

4.1 Calendar alignment and missing observations

Cleaning standardizes dates and types, removes identical duplicates, reviews conflicting records and isolates invalid prices before SPY-calendar alignment. Missing ETF prices and volumes remain missing; genuine extreme returns and provider precision are preserved. HYG's first observation is 11 April 2007, leaving 318 earlier sessions empty. Complete common features begin on 9 May 2007, before initial estimation starts. VIX excludes thirty-three non-SPY dates and uses closing values; its inconsistent 8 February 2006 opening quote is isolated.

DGS3MO missing markers become null; genuine zeros remain. A yield is assumed released on the next Federal Reserve publication day and usable after that day ends. Entry dates match the latest nonmissing yield released strictly before entry. Source dates are retained and ages above seven calendar days flagged. Cash accrual is yield/100 × elapsed calendar days/365. DGS3MO serves accounting only. Cleaning checks are exported by 01_download_clean_data.ipynb.

4.2 Cash returns and verification

Using closing price C in the same share units as O and D, historical cash return is

$$r_{i,t}^{\mathrm{cc}} = \frac{C_{i,t} + D_{i,t}}{C_{i,t-1}} - 1 \tag{3}$$

Independent reconstruction from opening prices and exit-date dividends reproduces all 46,943 valid targets: 46,962 date–sector records minus eighteen terminal labels and one XLF event-crossing label. Adjusted close cross-checks closing returns; adjusted OHLC and Yahoo JSON check provider adjustments and library processing. The intraday–overnight decomposition of open-to-open returns holds across 77,936 intervals, with maximum absolute residual 2.22×10^{-16}.

4.3 Corporate actions and dependency exclusions

XLF's real-estate separation distributed XLRE shares in kind, with an ex-date of 19 September 2016.[3] Yahoo records a split-like factor of 1.231. We exclude event-crossing returns and the label signalled on 15 September,

[3] Select Sector SPDR Trust, 8 June 2016 supplement; XLF, Form 8937 (signed 15 November 2016).

entering on 16 September and exiting on 19 September. Unaffected intraday observations and subsequent ordinary returns are retained.

Affected rolling features include XLF's own history, other sectors' XLF predictors and sector relationship summaries. XLF's dependency windows extend through 18 September 2017. For the other sectors, sixty S2 dates through 12 December 2016 are excluded; the common-sample rule also removes these dates from S0. Explicit reasons and dependency windows are retained in the cleaning and feature checks.

Five ordinary 2:1 splits—XLB, XLE, XLK, XLU and XLY—took effect before the open on 5 December 2025.[4] Audits retain provider price and volume units without a second adjustment.

4.4 Feature definitions and economic interpretation

Table 3: Feature definitions and economic roles

First set	Feature family and count	Definition and lookback	Economic interpretation
S0	Own return history (7)	Daily returns at t, t-1 and t-2; 5-, 20- and 60-day cumulative returns; momentum compounding 231 returns from t-251 to t-21	Short-term continuation or reversal and longer-horizon momentum
S0	SPY-relative history (4)	Own minus SPY cumulative returns over 1, 5, 20 and 60 days	Recent sector outperformance
S0	Trading sessions (4)	Latest overnight cash return and intraday return; five-day mean of each	Different information and trading activity across sessions
S0	Volatility (3)	20- and 60-day sample standard deviations; 20-day Parkinson high–low volatility	Return risk and within-session price variation
S0	Trading activity (2)	Current volume divided by the preceding twenty sessions' mean; log of the twenty-day mean of (close × volume)	Unusual activity and trading scale
S0	Illiquidity (1)	Twenty-day mean of the daily ratio of absolute cash return to dollar volume, scaled by 10^6	Price movement per unit of trading activity
S0	Market exposure (2)	Sixty-day covariance with SPY divided by SPY variance; sixty-day correlation with SPY	Sensitivity to and co-movement with the market
S0	SPY history (7)	Daily returns at t, t-1 and t-2; 5-, 20-, 60-day cumulative returns; twenty-day volatility	Common market direction and risk
S0	VIX (3)	Log closing level; one- and five-day percentage changes	Expected market uncertainty and its recent change
S0	Auxiliary ETFs (20)	For each of TLT, IEF, GLD, HYG and LQD: 1-, 5-, 20-day cumulative returns and twenty-day volatility	Bond, gold and credit-market conditions
S0	Return contrasts (4)	TLT minus IEF and HYG minus LQD cumulative returns over 5 and 20 days	Relative performance within Treasury and corporate-bond ETFs
S1	Other sectors (40)	For each of eight named sectors: daily returns at t, t-1 and t-2, plus 5- and 20-day cumulative returns	Candidate lead–lag information from other sectors
S2	Sector rank (1)	Rank of own twenty-day return among nine sectors, normalized as (rank-1)/8; average ranks for ties	Relative momentum position
S2	Sector dispersion (2)	Cross-sectional sample standard deviations of nine 1-day and nine 20-day returns	Divergence in sector performance
S2	Sector correlation (2)	Mean signed sixty-day correlation of target with other eight sectors; mean across all thirty-six unique sector pairs	Target integration and common sector co-movement

The dollar-volume proxy is close × volume. Overnight and intraday returns are

$$r_{i,t}^{\text{on}} = \frac{O_{i,t} + D_{i,t}}{C_{i,t-1}} - 1, \qquad r_{i,t}^{\text{id}} = \frac{C_{i,t}}{O_{i,t}} - 1 \tag{4}$$

Momentum follows Jegadeesh and Titman (1993); illiquidity follows Amihud (2002). The Parkinson (1980) estimator is

$$\sigma_{i,t}^{\text{PK}} = \sqrt{\frac{\text{mean}_{20}\left[\ln^2(H/L)\right]}{4\ln 2}} \tag{5}$$

[4] State Street, share-split announcement, 20 November 2025.

H and L are daily high and low prices; the mean covers twenty complete sessions. Together, these features describe sector return dynamics conditional on market risk, trading frictions and financing conditions. Exact return contrasts create collinearity: Elastic Net coefficients depend on the correlated representation and penalty, limiting attribution to individual columns.

4.5 Preprocessing and panel construction

Fixed feature formulas leave missing values unfilled. Scaling and any clipping are fitted within eligible monthly training windows; full-sample diagnostics are descriptive. The shared panel preserves the full calendar with separate targets, timestamps and eligibility flags, sorted uniquely by date and ticker. Twenty-seven CSVs contain ordered predictors for nine sectors and three specifications, excluding each target's own cross-sector columns. Feature definitions, dependencies and references are provided in 02_features_panel_data.ipynb.

4.6 Limitations

Data and execution: First, the 18:00 convention assumes daily Yahoo and Cboe observations are available; historical publication timestamps were not observed. Second, Yahoo and FRED provide revisable final snapshots, and Yahoo JSON/CSV agreement checks processing rather than independent data accuracy. Third, openings are execution proxies. Among valid sector-day gaps, those within 0.01 bp of zero account for 3.08% in training, 1.47% in validation and 1.43% in testing. We retain them; quote executability and trading frictions remain portfolio assumptions.

Sector definitions: First, reclassifications alter exposures: the 2016 XLF separation affects training, while the 2018 Communication Services expansion affects XLK and XLY in validation.[5] The revision effective after the close on 17 March 2023 moves payment processors, including Visa, Mastercard and PayPal, to Financials, other technology services to Industrials and selected retailers to Consumer Staples, affecting XLK, XLF, XLI, XLY and XLP in testing.[6] Results therefore reflect evolving exposures. Second, the nine-fund universe partly covers today's eleven sectors; SPY-relative targets retain the exposure properties in Section 2.1.

Implementation: notebooks 01–02; artifact paths: Appendix B.

5 Elastic Net

Building on the research design, sample split, and nested S0/S1/S2 feature specifications defined in the preceding sections, we first use Elastic Net as the linear forecasting benchmark. The model serves two purposes. First, it tests whether cross-sector information provides incremental predictive value within a regularized linear framework. Second, it provides a linear benchmark for the GBDT model in the next section using the same target and scoring sample. All Elastic Net forecasts are subsequently passed to the common portfolio backtest together with the GBDT forecasts, so this section focuses only on the forecasting model and does not repeat the later position, transaction-cost, or return accounting.

5.1 Model formulation

Elastic Net combines L1 and L2 regularization. For one sector, one feature specification, and one monthly training sample, let X denote the standardized predictors and y the next-period sector-minus-SPY return. The model solves

$$\min_{b,\beta} \frac{1}{2n}\|y - b - X\beta\|_2^2 + \alpha\rho\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2, \tag{6}$$

where α controls the overall strength of regularization, ρ (`l1_ratio`) controls the relative contribution of the L1 and L2 penalties, and the intercept b is unpenalized. The L1 component can shrink weak predictors exactly to zero, while the L2 component stabilizes estimation when predictors are highly correlated. This makes Elastic Net suitable for the correlated sector-return, market-state, and auxiliary-asset features constructed earlier (Zou and Hastie, 2005).

Separate models are estimated for the nine sector ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, and XLY. Consistent with the earlier design, the target is the dividend-inclusive open-to-next-open return

[5] S&P Dow Jones Indices, GICS overview, April 2026, p. 5.

[6] S&P DJI/MSCI, 31 March 2022 notice; State Street, Chesworth, February 2023, pp. 1–3.

of the sector ETF minus the corresponding SPY return. S0, S1, and S2 contain 57, 97, and 102 predictors, respectively. Because the feature definitions, data sources, and information timing have already been specified in the preceding data and feature-construction sections, this section uses the frozen feature files directly and does not redefine or reselect the feature sets.

5.2 Monthly training and preprocessing

Elastic Net follows the same monthly expanding-window forecasting protocol used throughout the project. The initial estimation history covers 2008–2017, 2018–2022 is used for forward validation and hyperparameter selection, and 2023 through September 2026 forms the final out-of-sample test period. Before each forecast month, the nine sector models are re-estimated at 18:00 New York time on the preceding SPY trading session. The fitted models remain fixed during that month, and the next monthly fit expands the training sample using newly matured labels.

The training data include only predictors available by the fitting cutoff and target labels that have already matured. Consistent with the label-maturity purge defined earlier, observations whose exit opening has not yet occurred are excluded from estimation. During testing, labels from earlier test months may enter later expanding-window fits once they become observable, but all hyperparameters remain fixed after validation.

Because Elastic Net is sensitive to variable scale, each monthly fit estimates a `StandardScaler` using only the contemporaneous training sample and then applies the same transformation to that month's forecast observations. The target remains in raw decimal-return units. No full-sample scaling, future-information imputation, or test-dependent preprocessing is used. The main scoring sample is shared with the other model family: 11,313 date–sector observations in validation and 8,433 in testing.

5.3 Validation grid and parameter selection

The hyperparameter grid is specified in advance as

$$\alpha \in \{10^{-5}, 10^{-4}, 10^{-3}\}, \qquad \rho \in \{0.2, 0.4, 0.6, 0.8\}.$$

Each feature specification therefore contains 12 parameter combinations, giving 36 scheme–parameter configurations across S0–S2. Within one specification, the nine sector models share the same (α, ρ) pair but estimate separate coefficients. Validation forecasts for 2018–2022 are generated under the same monthly expanding-window protocol used in testing, and squared errors are pooled across all months and sectors. The final selection rule minimizes pooled validation MSE. Exact MSE ties prefer the larger α, followed by the larger ρ.

Table 4: Validation-selected Elastic Net configurations

Scheme	Predictors	α	`l1_ratio` ρ	Validation MSE	Validation R^2_{OOS}
S0	57	0.001	0.8	9.925382×10^{-5}	-0.000691
S1	97	0.001	0.8	9.924534×10^{-5}	-0.000605
S2	102	0.001	0.8	9.924534×10^{-5}	-0.000605

All three specifications select `alpha = 0.001` and `l1_ratio = 0.8`. S1 reduces validation error only slightly relative to S0, while S1 and S2 are numerically almost identical. Because the research design explicitly compares S0, S1, and S2, all three specifications are retained for testing rather than selecting a single specification based on such small validation differences.

A second important result is that the selected parameters for all three specifications lie at the strongest-regularization corner of the current grid. This indicates that validation favors substantial coefficient shrinkage. We do not expand the α grid after observing test results, because doing so would redesign the model using final test information.

5.4 Forecast performance

Forecast evaluation uses the same historical-mean benchmark as the GBDT section. For each scored observation, the benchmark forecast is the historical mean of the same sector's target computed from the eligible training labels available at that month's refit. Out-of-sample R^2 is defined as

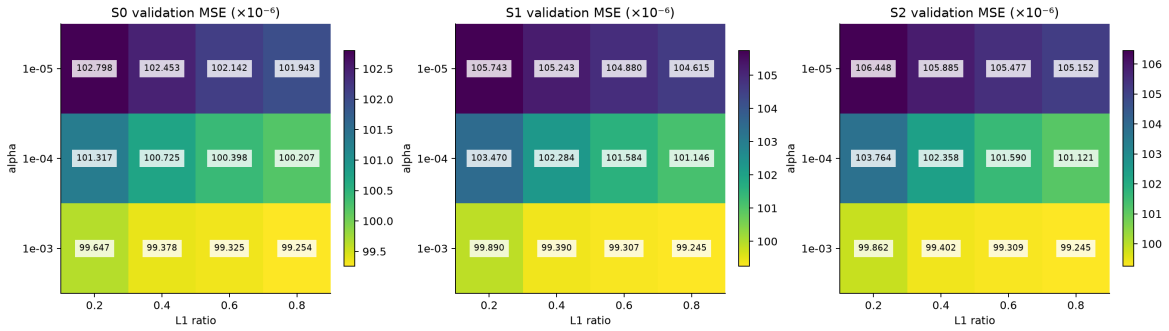


Figure 1: Elastic Net validation grid. The figure compares pooled validation MSE across the prespecified (α, ρ) grid for the three feature specifications. All three achieve their lowest validation MSE at $\alpha = 0.001, \rho = 0.8$

$$R^2_{\text{OOS}} = 1 - \frac{\sum(y - \hat{y})^2}{\sum(y - \bar{y}_{\text{hist}})^2}. \tag{7}$$

Positive values indicate lower squared error than the historical-mean benchmark, while negative values indicate underperformance relative to that benchmark. Direction accuracy measures agreement between the sign of the forecast and the realized sector-minus-SPY return.

Table 5: Elastic Net forecast performance

Sample	Scheme	Observations	RMSE	MAE	R^2_{OOS}	Direction accuracy
Validation	S0	11,313	0.009963	0.006759	-0.000691	0.504375
Validation	S1	11,313	0.009962	0.006759	-0.000605	0.504552
Validation	S2	11,313	0.009962	0.006759	-0.000605	0.504552
Test	S0	8,433	0.009566	0.006932	-0.000473	0.508241
Test	S1	8,433	0.009566	0.006932	-0.000473	0.508241
Test	S2	8,433	0.009566	0.006932	-0.000473	0.508241

During validation, S1 and S2 have slightly higher R^2_{OOS} than S0, but all three remain below the historical-mean benchmark. In the test period, S0, S1, and S2 produce identical predictions observation by observation, and therefore have the same RMSE, MAE, R^2_{OOS}, and direction accuracy. The pooled test R^2_{OOS} is -0.000473, indicating slightly higher overall squared error than the historical mean, while direction accuracy is 0.508241. This result is important for the subsequent model comparison. Because the three Elastic Net specifications generate identical test forecasts, they also generate identical test-period trading paths and portfolio results in the downstream backtest. The next GBDT section uses the same target and scoring sample, so differences between GBDT and Elastic Net can be attributed to model form rather than sample coverage.

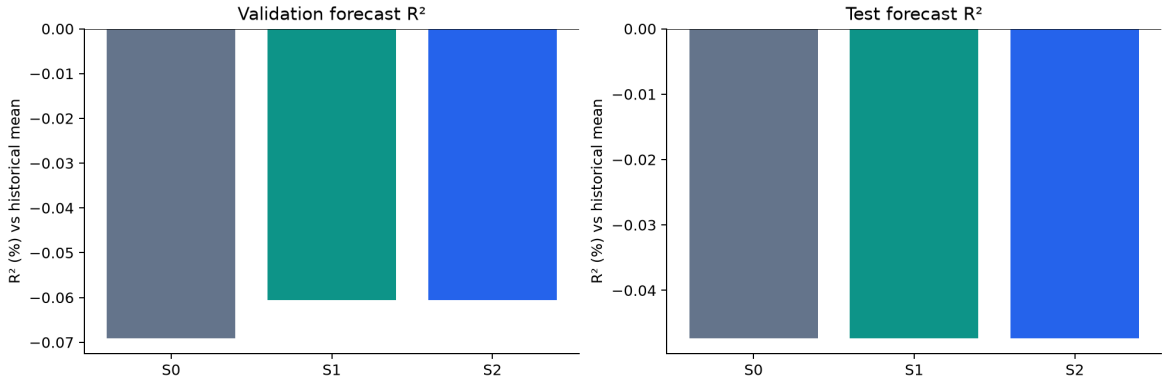


Figure 2: Elastic Net out-of-sample forecast performance. The figure compares R^2_{OOS} relative to the historical mean for S0, S1, and S2 in validation and testing. S1/S2 improve only slightly during validation, while all three specifications overlap exactly in the test period

5.5 Sparsity and feature selection

The test period contains 45 forecast months. With nine sectors estimated separately, each specification therefore contains 405 sector–month fits. The locked Elastic Net models are highly sparse.

Table 6: Test-period coefficient sparsity

Scheme	Sector–month fits	Mean nonzero coefficients	Median	Maximum	Intercept-only fits
S0	405	0.348	0	3	87.41%
S1	405	0.348	0	3	87.41%
S2	405	0.348	0	3	87.41%

In 87.41% of sector–month fits, every predictor coefficient is zero and the model retains only the intercept. More importantly, the `cross_` variables added in S1 and the `rel_` relationship summaries added in S2 have zero selection frequency during testing. Although S1 and S2 therefore contain larger candidate feature sets, all newly added variables are removed by the locked Elastic Net models. This directly explains why the three specifications generate identical test forecasts.

Only a small number of variables remain active during testing. `own_intraday_1d`, `own_excess_SPY_5d`, and `aux_LQD_cumret_5d` each appear in 11.11% of sector–month fits, while `aux_HYG_cumret_1d` appears in 1.48%. These coefficients are concentrated in only a few sectors and should be interpreted as forecasting selections rather than causal relationships.

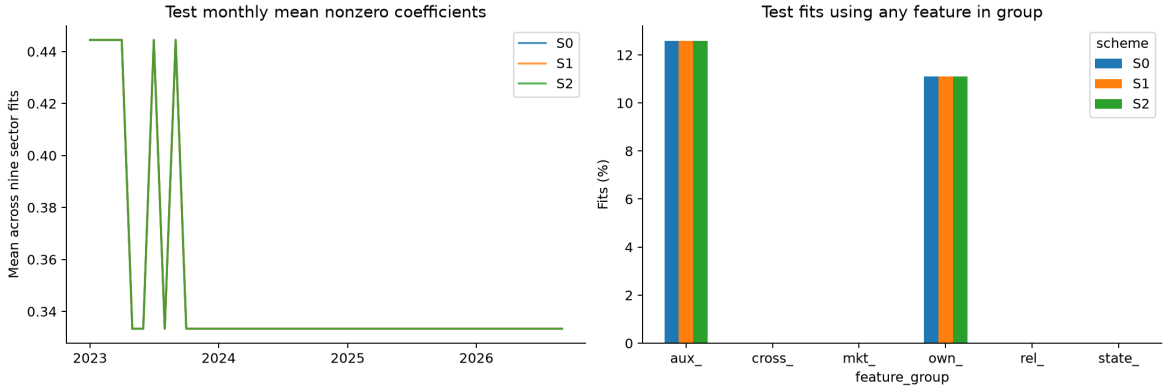


Figure 3: Elastic Net sparsity and feature-group diagnostics. Most test-period fits collapse to intercept-only models, while the cross-sector and relationship-summary feature groups do not enter the final forecasts

5.6 Sector heterogeneity

Although the pooled test R^2_{OOS} is only slightly negative, the forecast error is not evenly distributed across the nine sectors. Because all three specifications produce identical test forecasts, Table 7 reports one common set of sector-level results.

Table 7: Test-period Elastic Net performance by sector

Sector	R^2_{OOS}	Sector	R^2_{OOS}	Sector	R^2_{OOS}
XLB	0.000000	XLE	0.000000	XLF	-0.008133
XLI	0.000000	XLK	0.000000	XLP	0.000000
XLU	0.000000	XLV	0.000000	XLY	-0.000003

XLF is the dominant negative contributor, with a test R^2_{OOS} of -0.008133. A decomposition shows that XLF accounts for approximately **99.96%** of positive sector excess SSE, defined as the amount by which model SSE exceeds historical-mean SSE. Coefficient selection is also highly concentrated in XLF: `own_intraday_1d`, `own_excess_SPY_5d`, and `aux_LQD_cumret_5d` are selected in all 45 XLF test months. By contrast, most other sectors are regularized to predictions that are almost identical to their historical means.

The aggregate negative result is therefore not driven by persistent poor forecasts across all nine sectors. Most sectors make almost no active forecast adjustment, while the small number of nonzero predictions is

concentrated mainly in XLF and does not improve test-period squared error.

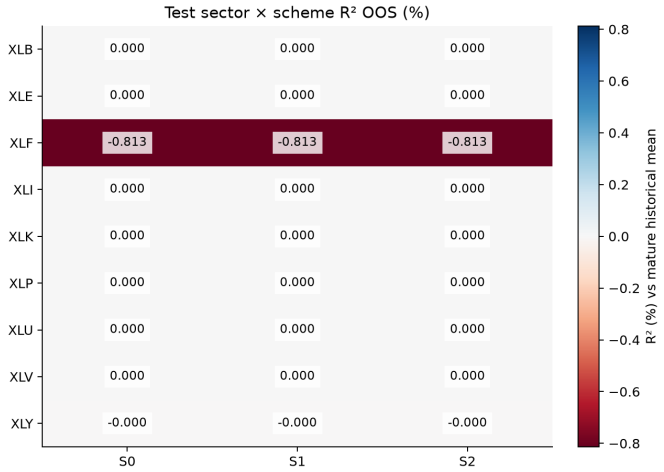


Figure 4: Sector-level Elastic Net test R^2_{OOS}. Because all three specifications produce identical test forecasts, the heatmap shows the same sector-level pattern for S0, S1, and S2. The negative forecast error is concentrated primarily in XLF

5.7 Robustness and diagnostic analysis

The main results show that Elastic Net does not outperform the historical mean in the test period and that S0, S1, and S2 generate identical test forecasts. We therefore conduct additional diagnostics focusing on the validation design, training-window length, regularization, sector concentration, and time variation. Alternative model and window comparisons use only the 2018–2022 validation period. Test-period XLF, rolling-loss, and HAC results are used only to describe the behavior of the already locked forecasts and do not change model selection.

5.7.1 Validation and training-window robustness

The main procedure selects parameters using pooled MSE, which gives greater influence to higher-volatility sectors in the squared-error objective. As a robustness check, we also use a sector-equal criterion that first computes losses within each sector and then gives all nine sectors equal weight. Across 18 comparisons covering Elastic Net and Ridge, three training windows, and three feature specifications, pooled MSE and the sector-equal criterion select exactly the same parameters. Parameter selection is therefore not sensitive to the way validation losses are aggregated.

We then compare expanding, rolling 5-year, and rolling 3-year training windows. Elastic Net uses the original 12 (α, ρ) combinations, while Ridge uses $\lambda \in \{0.01, 0.1, 1, 10\}$. The full diagnostic contains 144 configurations and 77,760 monthly-sector fits.

The complete comparison is reported in Table 18 in Appendix A.

All selected Elastic Net configurations use $\alpha = 0.001, \rho = 0.8$, while all selected Ridge configurations use $\lambda = 10$.

Shorter training windows do not improve Elastic Net. For all three schemes, the rolling 5-year and rolling 3-year results are weaker than the expanding-window results, with the 3-year window performing worst. Ridge shows a similar pattern: expanding-window performance is slightly positive, while the rolling-window versions are around zero or below. The weak Elastic Net performance therefore cannot be explained simply by an overly long training history.

5.7.2 Regularization and sector concentration

Under expanding-window validation, the selected Elastic Net models contain on average only 0.557, 0.756, and 0.756 nonzero coefficients for S0, S1, and S2, respectively. The corresponding intercept-only fit rates are approximately 72.41%, 67.78%, and 67.78%.

Ridge retains all predictors and achieves validation R^2 values of +0.1133%, +0.1288%, and +0.1307% under the same expanding-window design, while Elastic Net remains slightly negative. This suggests that the selected Elastic Net penalty may remove weak but dispersed linear information too aggressively.

At the same time, Elastic Net's negative test error is highly concentrated in XLF. XLF has a test R^2_{OOS}

of -0.813%, accounts for **99.96%** of positive sector excess SSE, and contributes **95.74%** of all nonzero coefficient-fit occurrences. Most other sectors are effectively close to historical-mean forecasts.

The complete comparison is reported in Table 19 in Appendix A.

The 5-year window improves XLF performance in some cases, but the 3-year window deteriorates again. The XLF result is therefore better characterized as unstable predictive performance than as a simple consequence of using too much historical data.

5.7.3 Time variation and diagnostic summary

Trailing-12-complete-month loss improvement shows clear time variation. Positive values indicate lower model MSE than the historical-mean benchmark. Elastic Net underperforms the benchmark during much of both validation and testing, but shows temporary improvements around 2021 and during part of 2026. S0, S1, and S2 overlap exactly in the test period because their forecasts are identical observation by observation.

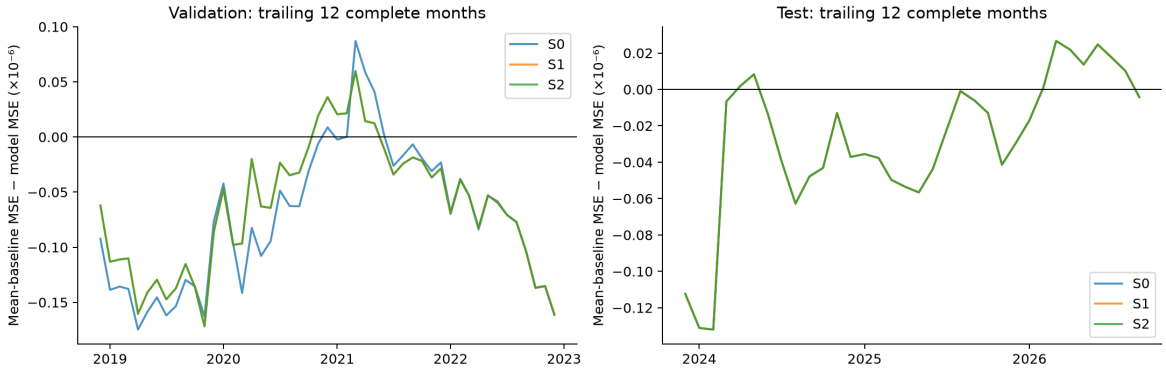


Figure 5: Rolling twelve-month loss improvement relative to the historical-mean benchmark. Positive values indicate lower model MSE than the historical mean; the three specifications overlap in testing because their forecasts are identical

Using Newey–West HAC standard errors with 20 trading-day lags, the test-period mean daily loss improvement of Elastic Net relative to the historical mean is

$$-4.33 \times 10^{-8},$$

with an approximate 95% interval of

$$[-1.14 \times 10^{-7},\ 2.72 \times 10^{-8}].$$

During validation, the loss improvement of S1 relative to S0 is only 8.48×10^{-9}, while the S2–S1 difference is close to machine precision. The differences among the three Elastic Net specifications are therefore very small and unstable over time.

Taken together, the diagnostics highlight three main features. First, strong regularization produces many intercept-only models and removes the additional cross-sector and relationship variables. Second, most of the negative test error is concentrated in XLF rather than spread evenly across sectors. Third, forecast performance varies substantially over time, while shorter training windows do not systematically improve the results. At the same time, pooled and sector-equal criteria consistently select the same parameters, so the validation aggregation rule is not a major driver.

Overall, Elastic Net appears limited by a highly sparse linear representation that has difficulty retaining weak, dispersed, and time-varying cross-sector information. This provides a direct motivation for the next GBDT section, where nonlinearities, threshold effects, and interactions can be modeled explicitly.

5.8 Conclusions and connection to GBDT

H1 Cross-sector information. Within the linear Elastic Net framework, the test results do not support the hypothesis that adding other sectors' return histories improves prediction. S1 produces only a very small validation MSE improvement over S0, while the two specifications generate identical test forecasts. None of the 40 additional cross-sector variables is retained by the locked test models. Cross-sector information therefore does not add measurable predictive value within this Elastic Net specification.

H2 Relationship representations. S2 is almost identical to S1 during validation and produces exactly the same test forecasts. The five additional rank, dispersion, and correlation summaries are all shrunk to zero during testing. These relationship representations therefore provide no incremental predictive value under Elastic Net.

These results also provide a clear benchmark for **H3 Nonlinearity**. Strong Elastic Net shrinkage pushes most models toward the historical mean and prevents the additional cross-sector features from entering the test forecasts. The diagnostics further show that changing the training window or validation aggregation rule does not resolve the pattern, while the small improvement from expanding-window Ridge suggests that the sparse linear restriction may be too strong. The next section therefore estimates Gradient Boosted Decision Trees using the same S0/S1/S2 specifications, timing protocol, and scoring sample to test whether nonlinearities and interactions can extract predictive structure that Elastic Net does not retain.

The alternative model, validation-criterion, and training-window comparisons in this section are conducted using validation data only, while the test-period XLF, rolling-loss, and HAC analyses describe the behavior of the already locked forecasts. None of these diagnostics changes the original model selection or regenerates test predictions. We do not expand the Elastic Net grid after observing test performance, remove XLF or earlier years, or replace the original model ex post with Ridge. The three original Elastic Net forecast series are preserved and passed unchanged to the common downstream backtest. This allows the portfolio section to compare the economic value of Elastic Net and GBDT under the same trading rules, thresholds, and cost assumptions while preserving the integrity of the test evaluation.

Implementation: notebooks 03; artifact paths: Appendix B.

6 Gradient Boosted Decision Trees

We developed a GBDT forecasting workflow for nine US sector ETFs under three nested feature specifications. The implementation combines monthly expanding-window estimation, a search over 81 scheme–parameter configurations, and validation-based selection of boosting rounds. We evaluated the forecasts against the historical mean and Elastic Net, examined performance by year and sector, and measured feature-group importance. Validation selected S1 with 43 boosting rounds; its validation and test out-of-sample R^2 values are 0.003587 and 0.000429, respectively.

6.1 Model formulation

GBDT constructs an additive prediction function by fitting small regression trees sequentially to the remaining error (Friedman, 2001). For one sector, one feature specification and one monthly training sample, let x_i denote the predictor vector and y_i the sector-minus-SPY target. Under squared-error loss, the initial prediction is the training mean and subsequent trees learn residual structure:

$$F_0(x) = \bar{y}_{\text{train}}, \qquad r_{im} = y_i - F_{m-1}(x_i) \tag{8}$$

$$F_m(x) = F_{m-1}(x) + \eta h_m(x), \qquad m = 1, \ldots, M \tag{9}$$

Here h_m is a regression tree, η is the learning rate and M is the selected number of boosting rounds. A tree divides observations into regions defined by predictor thresholds. Successive splits allow a return signal to have different effects under different market conditions, without specifying those interactions in advance.

6.2 Leaf regularization and complexity

The implementation uses HistGradientBoostingRegressor with squared-error loss. It groups continuous predictors into histogram bins before searching for splits. For a fixed leaf region R_{jm}, the unscaled squared-loss leaf update can be expressed as

$$\gamma_{jm} = \frac{\sum_{i \in R_{jm}} r_{im}}{n_{jm} + \lambda} \tag{10}$$

The leaf sample count n_{jm} and L2 penalty λ control the size of the update; the learning rate then scales its contribution. Maximum leaves and depth restrict interactions, while a minimum leaf size limits highly local fits. L2 regularization shrinks leaf predictions toward zero.

6.3 Forecast target and feature specifications

Signals use information through the close at 18:00 New York time. The target is the next-open-to-following-open cash return of a sector ETF minus the corresponding SPY return. Separate models are estimated for XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV and XLY. Returns and predictions are expressed in decimal units.

Table 8: Nested GBDT feature specifications

Scheme	Predictors	Information added
S0	57	Own-sector features and common market controls
S1	97	S0 plus 40 return-history features from the other eight sectors
S2	102	S1 plus five sector rank, dispersion and correlation summaries

6.4 Monthly training and parameter selection

Initial estimation uses eligible history from 2008–2017. Validation forecasts cover 2018–2022, followed by test forecasts for 2023–September 2026. Before each forecast month, every sector model is refitted using expanding history from 2008. Features, signals and matured target labels must be available by 18:00 on the preceding SPY trading session. The fitted trees remain fixed within that month.

This is a forward forecasting exercise: the model predicting January 2018 uses only information available before that month. Once their labels mature, January observations may enter February estimation. The same updating rule continues during testing. Hyperparameters stay fixed after validation selection, while tree splits and leaf values are re-estimated each month. Predictors are used in their original units, without scaling, imputation, or winsorization.

Table 9: Search space and fixed settings

Parameter	Candidates or setting
Maximum leaves	3, 5, 7
Minimum samples per leaf	10, 20, 50
L2 leaf penalty	0, 1, 10
Boosting rounds	1–50; patience 10; MSE tolerance 10^{-8}
Learning rate and maximum depth	0.03 and 4
Histogram bins and random seed	127 and 5205

6.5 Pooled validation and round selection

The grid contains 27 structural configurations per scheme, or 81 across S0–S2. For every configuration, each monthly sector model is fitted once through 50 rounds. Staged predictions provide the validation error at every round. Squared errors and observation counts are summed across months and sectors, so selection minimizes pooled MSE rather than the average of sector R^2 values.

A patience rule scans each completed loss curve. An improvement greater than 10^{-8} resets the counter relative to the last significant best MSE; ten rounds without such improvement end the scan. The lowest actual MSE visited is retained. The rule selects the model length from the precomputed validation path. Exact ties prefer fewer leaves, fewer rounds, larger L2 and larger minimum leaf size. Selection uses the existing 2018–2022 validation period.

Table 10: Validation-selected configurations

Scheme	Max leaves	Min leaf	L2	Best round	Stop round
S0	5	20	10	23	33
S1	3	20	10	43	48
S2	3	20	10	40	46

S1 has the lowest pooled validation MSE and is the primary specification. Its 43-tree model uses three leaves per tree and L2 = 10. S0 and S2 are retained for feature comparisons. The selected rounds are 23, 43 and 40, with patience stopping points of 33, 48 and 46, respectively.

6.6 Forecast performance

Forecast evaluation uses a common sample across all three specifications: 11,313 validation observations and 8,433 test observations, corresponding to 1,257 and 937 dates across nine sectors. The benchmark for each observation is the same sector's historical target mean, calculated from the eligible training sample at that month's refit. The out-of-sample measure is

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{i,t}(y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t}(y_{i,t} - \bar{y}_{i,\text{train}(t)})^2} \tag{11}$$

Positive values indicate lower squared error than the historical mean. Because the statistic pools squared errors, high-volatility observations contribute more to the comparison. Direction accuracy measures agreement between the signs of the forecast and the realized SPY-relative return.

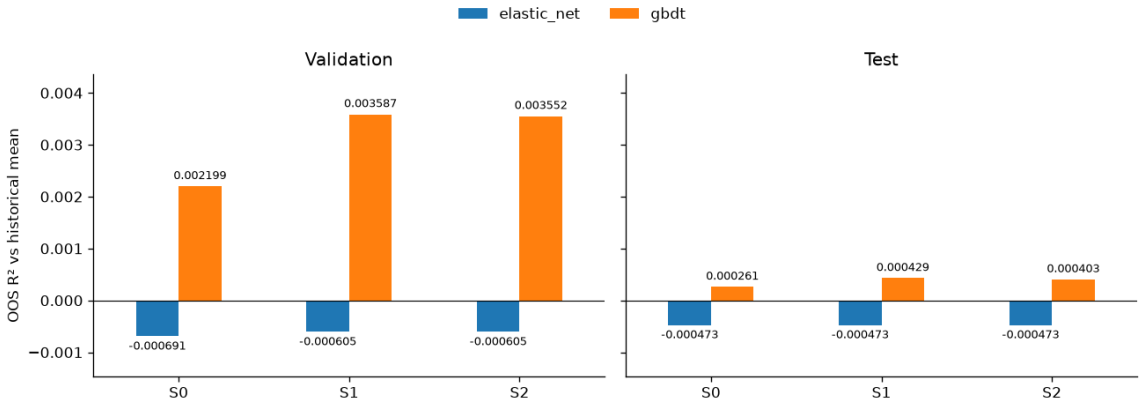


Figure 6: Model comparison across feature specifications

For S1, test RMSE is 0.009562 and direction accuracy is 0.497569. Relative to the original Elastic Net S1 benchmark, test R^2 rises from -0.000473 to 0.000429 and RMSE falls from 0.009566 to 0.009562. Both models use the same target and scoring sample. This is a small observed advantage for the fitted GBDT procedure. The comparison reflects combined differences in nonlinearity, regularization, model capacity and fitting procedures.

For S1, mean test squared-loss improvements are 3.92×10^{-8} against the historical mean and 8.25×10^{-8} against Elastic Net. Approximate 95% Newey-West intervals (20 lags), based on sector-averaged daily loss differences, are $[-1.83 \times 10^{-7}, 2.61 \times 10^{-7}]$ and $[-1.55 \times 10^{-7}, 3.20 \times 10^{-7}]$, respectively. Both intervals include zero, indicating uncertainty around these small positive estimates.

6.7 Stability across years and sectors

We evaluated each specification by calendar year and sector to identify the sources of its aggregate performance. Figure 7 compares S0, S1 and S2 over 2018–2026, with a dashed line separating validation and testing. Its vertical axis reports R^2 in percent; for example, 0.1% corresponds to 0.001 in decimal units. The 2026 results cover the available period through September.

In validation, 2020 and 2022 together account for approximately 94.5% of S1's net reduction in squared error against the historical mean. Their influence is amplified by the pooled squared-error objective. The model underperforms the mean in 2019 and 2021 despite a positive aggregate validation score. The gains are concentrated in particular years.

The test period has positive S1 R^2 in 2023 and 2025, and negative R^2 in 2024 and 2026. The strongest test year is 2025 at 0.003241. These sign changes indicate substantial variation in predictive performance over time.

Only XLK, XLP and XLU beat their own historical-mean forecasts in the S1 test results. XLB reverses from validation R^2 of 0.013305 to test R^2 of -0.006290. The aggregate gain is concentrated in three sectors, with contributions weighted by their benchmark squared errors.

The annual and sector evaluations are exported alongside the aggregate metrics. They show that S1's positive aggregate result combines gains and losses across different years and sectors. The overall ranking therefore describes average performance in this sample rather than a uniform or stable improvement across market

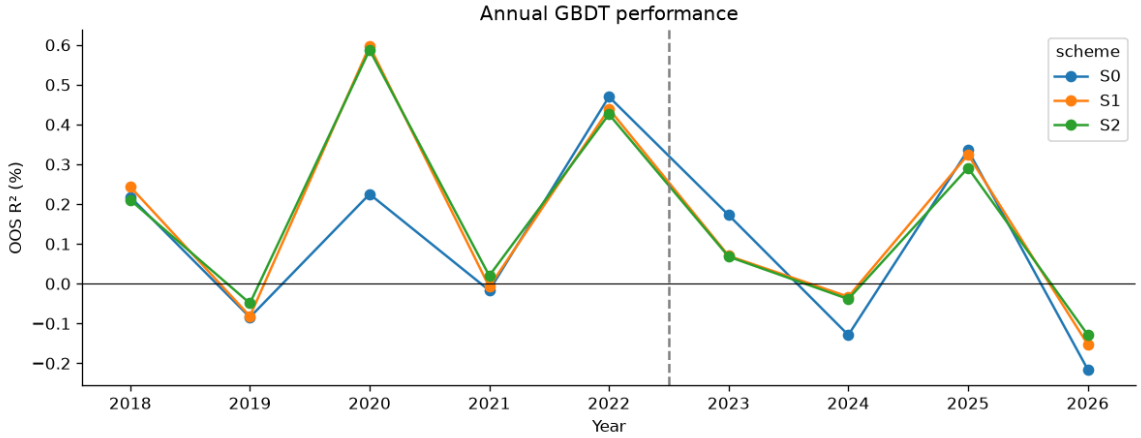


Figure 7: Annual GBDT forecast performance

Table 11: S1 test performance by sector

Sector	R^2 OOS	Sector	R^2 OOS	Sector	R^2 OOS
XLB	-0.006290	XLE	-0.001297	XLF	-0.001839
XLI	-0.003295	XLK	0.002727	XLP	0.004546
XLU	0.005395	XLV	-0.001365	XLY	-0.000242

conditions and industries.

6.8 Feature interpretation

Feature-group importance is measured on test observations using the monthly fitted models. Within each sector-month, all columns in one group are jointly shuffled using the same row permutation. Predictions are recomputed without refitting, and the change in squared error is averaged over three permutations before pooling across observations. A positive MSE increase indicates reliance on that group.

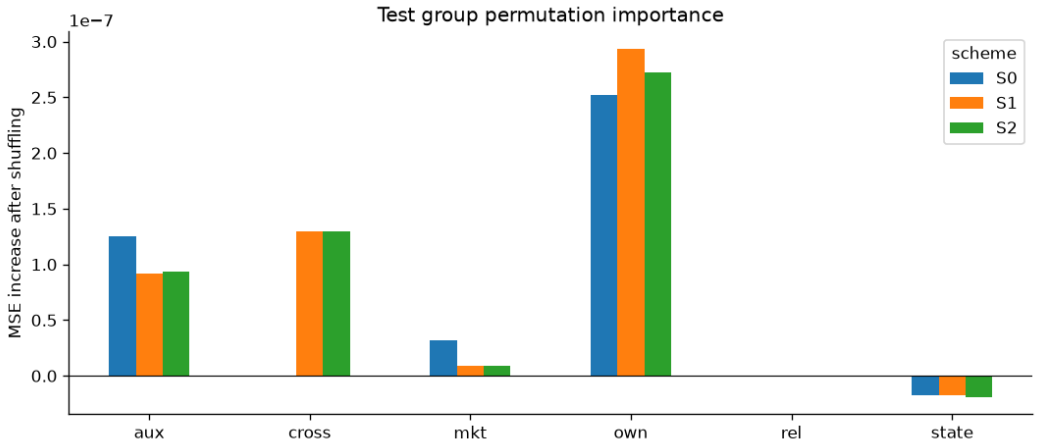


Figure 8: Feature group permutation importance

6.9 Conclusions

H1 Cross-sector information: S1 has higher aggregate R^2 point estimates than S0 in validation and testing. This is consistent with useful information in other sectors' return histories, and permutation importance shows that the fitted models use these features. The observed gains are small and vary across years and sectors. These results suggest that cross-sector histories offer potential predictive value within the evaluated sample.

H2 Relationship summaries: S1's aggregate point estimates are slightly better than S2's in both periods. The five rank, dispersion and correlation summaries provide no observed incremental improvement in this

evaluation, consistent with their near-zero permutation importance. Together, these findings favor retaining detailed cross-sector histories in S1 under the validation criterion used in this study.

H3 Nonlinearity: GBDT has higher aggregate out-of-sample R^2 point estimates than Elastic Net across S0, S1 and S2 in both periods. However, S1's test loss-improvement interval includes zero. This model comparison therefore suggests a small predictive advantage, without establishing a reliable advantage or identifying nonlinearity as its only cause. Regularization, capacity and fitting also differ.

Taken together, the results favor a regularized GBDT model with cross-sector return histories. The selected S1 specification combines the benefits of additional sector information and nonlinear prediction, while the loss-improvement intervals include zero and performance differs by year and sector. We therefore retain S1's monthly forecasts as a validation-selected input for evaluating investment performance in downstream backtesting.

Implementation: notebooks 04; artifact paths: Appendix B.

7 Portfolio Construction and Backtesting Results

7.1 Portfolio Evaluation Framework

7.1.1 From Forecast Errors to Trading Returns

This section translates the Elastic Net and GBDT forecasts under S0, S1 and S2 into trading portfolios. A common asset universe, execution schedule and cost framework allow comparison of their economic value, incorporating signal strength, trading frequency and position retention.

We evaluate two portfolio structures. The cash-based strategy initiates sector–SPY pairs when forecasts satisfy the trading thresholds. The SPY overlay maintains market investment and adjusts sector allocations through long and short positions. Their respective benchmarks are cash and SPY with dividend reinvestment, reflecting the different market exposures of the two structures.

7.1.2 Evaluation Periods and Model Parameters

Following the preceding sample division, validation spans the opening on 3 January 2018 to the opening on 30 December 2022, comprising 1,257 holding intervals. Testing spans the opening on 4 January 2023 to the opening on 30 September 2026, comprising 937 intervals. Each configuration starts each period independently with capital of 1 and no positions, and liquidates at the final opening.

Model hyperparameters were selected by the lowest validation mean squared error within each feature specification before forecasts were supplied to the backtest. The model sections provide the search grids and estimation procedures. For GBDT S1, validation selected three maximum leaf nodes, 20 minimum samples per leaf, L2 regularization of 10 and 43 boosting rounds. Test-period performance did not enter this hyperparameter selection.

During testing, hyperparameters remain fixed while models are refitted monthly using expanding windows and labels available by the fitting cutoff. These training samples include earlier test-period observations once their outcomes have been realized. Thus, scheduled model refitting uses newly available historical data under the previously selected hyperparameters.

7.1.3 A Common Threshold for the Original Pair Strategy

Model parameters and trading thresholds are determined separately. To compare the six forecast specifications under common trading conditions, the original pair strategy uses a shared entry threshold h_{in}, with the exit threshold fixed at $h_{\text{out}} = h_{\text{in}}/2$. Entry candidates are 5, 10, 15, 20 and 30 bp, where 1 bp = 0.0001. Each candidate is applied to all six strategies over the full validation period, producing 30 candidate portfolio paths. The selection criterion is the average daily return in excess of cash across the six strategies:

$$Q(h) = \frac{1}{6}\sum_{s=1}^{6}\left[\frac{1}{N_V}\sum_{k=1}^{N_V}\left(r_{s,k}^{\text{net}}(h) - r_k^f\right)\right], \tag{12}$$

where $N_V = 1{,}257$, $r_{s,k}^{\text{net}}(h)$ is the net holding-interval return of strategy s under threshold h, and r_k^f is the cash return over the same interval. The calculation retains all intervals without active positions and includes entry,

exit and holding costs. Candidates are ranked using unrounded objective values; exact ties favor the higher threshold.

The 15 bp candidate maximizes this validation objective (Table 12). Entry/exit thresholds of 15/7.5 bp were therefore fixed on 31 October 2026, before the original pair strategy's test-period trading evaluation.

Table 12: Validation-period threshold comparison for the original pair strategy

Entry threshold bp	Exit threshold bp	Mean daily return above cash bp	Mean participation rate %	Mean annualized two-sided turnover (times)
5	2.5	0.2278	57.995	38.759
10	5	0.2473	29.581	22.084
15	7.5	0.4059	13.683	11.263
20	10	0.2420	8.353	6.763
30	15	0.1366	4.031	2.948

Note: The last three columns are equally weighted averages across the six model–specification combinations. Participation is the proportion of intervals with an active sector pair at the start of the interval. Turnover is based on actual net traded notional.

7.1.4 Scope of the SPY Overlay Exploration

After examining test-period forecast evaluations and the original pair strategy's results, we introduced a SPY core holding to examine whether sector adjustments could add net returns to continuous market investment.

On 1 November 2026, we compared entry/exit thresholds of 10/5 bp and 15/7.5 bp across two models and three feature specifications, giving 12 overlay configurations. The lower thresholds examined greater sector participation; 15/7.5 bp retained the original reference. Validation comparisons preceded test-period evaluation, using existing forecasts and the same nine-sector universe.

After inspecting all 12 test results, we selected GBDT S1 with 15/7.5 bp thresholds on the same date because it had the highest test-period net annualized return. Both the introduction of the overlay and the final strategy selection used observed test information; these results are reported as ex post exploration. This strategy selection was separate from the earlier validation-MSE selection of model hyperparameters.

7.2 Trading Signals and Portfolio Construction

7.2.1 Signal Timing and Position Retention

Following the information-timing convention established earlier, signals are formed at 18:00 New York time on trading day t and executed at the next SPY session's opening. The forecast $\hat{y}_{i,t}$ predicts sector i's dividend-inclusive return from the opening at $t + 1$ to the opening at $t + 2$, less SPY's corresponding return. The forecast horizon is one open-to-open interval, while positions may continue across several intervals when successive signals satisfy the retention rule.

Trading decisions first examine existing pairs. Let $d_i \in \{-1, +1\}$ denote the position direction: +1 represents a long sector position paired with a short SPY position, and -1 represents the reverse. The existing direction is retained only when

$$d_i \hat{y}_{i,t} > h_{\text{out}}, \tag{13}$$

and is closed otherwise. An existing pair may remain open after falling outside the day's top four forecast ranks, provided that it satisfies the retention condition.

New entries are restricted to the four largest absolute forecasts among the nine sectors. Entry requires $|\hat{y}_{i,t}| > h_{\text{in}}$, subject to a maximum of four active pairs. Retained positions occupy capacity first. Forecast ties follow a fixed ETF order, and sectors below the top four are not used to fill remaining capacity. A reversal closes the existing direction and opens the opposite direction only if all entry conditions are satisfied. Threshold comparisons are strict, signals are reviewed daily, and there is no minimum holding period.

Trading eligibility is determined by the features available at the time, complete forecasts for all nine sectors, the model month and the fitting cutoff. An incomplete signal causes the original strategy to close its pairs at the next opening and the overlay strategy to restore a pure SPY allocation. Missing execution prices terminate the backtest. Intervals without trades and intervals with incomplete signals remain in the portfolio return series.

7.2.2 Cash-Based Sector–SPY Pairs

In the original strategy, each active pair allocates ±12.5% of pre-trade net asset value to its sector and SPY legs. For sector direction d_i, the target sector weight is $0.125d_i$, with a corresponding SPY weight of $-0.125d_i$. A maximum of four pairs is permitted, and unused capacity receives no additional allocation.

The SPY legs are aggregated into a single net position, and trades are determined from actual changes in shares. The portfolio restores its target weights daily. When no pairs are active, it holds cash and earns the corresponding cash return.

7.2.3 A SPY Core Holding with Sector Overlays

The overlay strategy uses the same sector directions and retention rules, with SPY as the allocation when no sector adjustments are active. Let $u_i = 0.125d_i$ denote the unscaled sector adjustment, with $u_i = 0$ when no active position is assigned to sector i. The aggregate unscaled sector short weight is

$$B_{\text{raw}} = \sum_i \max(-u_i, 0). \tag{14}$$

When $B_{\text{raw}} > 0$, the common scaling factor is

$$\lambda = \min\left(1, \frac{0.25}{B_{\text{raw}}}\right); \tag{15}$$

otherwise, $\lambda = 1$. Final weights are

$$w_i = \lambda u_i, \qquad w_{\text{SPY}} = 1 - \sum_i w_i. \tag{16}$$

The scaling preserves relative sector allocations while limiting sector short exposure to 25%. Post-trade net weights sum to 100% and gross exposure is capped at 150%. Without sector adjustments, the portfolio holds 100% SPY.

The overlay strategy restores its target weights daily, permits fractional shares and funds execution fees within the allocation budget. Let V^- denote pre-trade net asset value, A_j^- the value of existing holdings in asset j at the current opening, and c the one-way execution fee rate. Post-fee target net asset value V^+ satisfies

$$V^+ + c \sum_j \left| w_j V^+ - A_j^- \right| = V^-. \tag{17}$$

Target asset values and net share trades follow from this equation. Pair-strategy sizing uses pre-trade net asset value; overlay sizing uses post-fee net asset value.

7.3 Cost Accounting and Evaluation Metrics

7.3.1 Execution and Holding Costs

The main setting charges a one-way execution fee of 3 bp on actual net traded notional. Fees apply to initial entry, daily adjustments and final liquidation. Actual net short positions incur a stock borrowing fee of 1% per year. Short-sale proceeds are restricted and receive no interest rebate.

For the overlay strategy, let C denote the cash account balance and B the collateral requirement for actual net short positions at the start of an interval. Interest-earning cash is $\max(C - B, 0)$, and the funding shortfall is $\max(B - C, 0)$. The former earns the interval cash return; the latter incurs financing interest at the available DGS3MO rate plus one percentage point. Financing interest and stock borrowing fees are accrued separately to represent collateral funding and securities borrowing costs.

Rates use the most recent nonmissing observation available before each interval under the assumed publication calendar. Accrual follows actual calendar days divided by 365, including weekends and holidays, consistent with the earlier data-availability rules.

7.3.2 Net Asset Value Accounting and Investment Benchmarks

Portfolio accounting uses the provider's split-normalized, dividend-unadjusted opening prices and cash dividends in matching share units. Existing positions are settled for opening-price changes, ex-dividend entitlements and holding costs before new trades are executed. Long shares held before the opening receive dividends, while short shares pay them. Changes in portfolio net asset value consist of price gains and losses, dividends, cash interest, stock borrowing fees, financing interest and execution fees.

The cash benchmark compounds available rates over the same holding intervals. Fixed-share SPY maintains its initial share count, retaining dividends in interest-earning cash; dividend-reinvestment SPY reinvests available dividends. Both use the strategy's initial and final openings and execution fee rate. Dividend-reinvestment SPY is the overlay's primary benchmark because both reinvest available funds. Fixed-share SPY also supports comparison with the original pair strategy.

7.3.3 Return, Risk and Trading Activity

For an evaluation period containing N holding intervals, with initial capital of 1 and terminal net asset value V_N, net annualized return is calculated as

$$\mathrm{CAGR} = V_N^{252/N} - 1. \tag{18}$$

Annualized volatility is the sample standard deviation of interval net returns multiplied by $\sqrt{252}$. The pair strategy's cash-excess Sharpe ratio divides 252 times mean daily return above cash by the annualized volatility of those excess returns. The cash benchmark's Sharpe ratio is undefined because its excess-return variance is zero.

Let Δr_k denote the overlay strategy's daily net return less that of SPY with dividend reinvestment. Annualized tracking error and the information ratio are

$$\mathrm{TE} = \mathrm{sd}(\Delta r_k)\sqrt{252}, \qquad \mathrm{IR} = \frac{252\,\overline{\Delta r}}{\mathrm{TE}}. \tag{19}$$

Maximum drawdown is calculated from the net asset value path, including initial capital of 1 and the initial execution fee. Two-sided turnover sums actual net traded notional at each opening divided by pre-trade net asset value and is annualized by $252/N$. Sector participation is the proportion of intervals beginning with a sector position; the overlay strategy's SPY core holding is excluded. Mean sector holding duration is the equally weighted average across continuous episodes in the same ETF and direction. Changes in position size do not end an episode; reversals and final liquidation do.

7.4 Results for the Original Pair Strategy

7.4.1 Returns and Risk under the Main Cost Setting

Table 13: Performance of the original pair strategy and investment benchmarks

Strategy or benchmark	Validation net annualized return %	Test net annualized return %	Test annualized volatility %	Test cash-excess Sharpe ratio	Test maximum drawdown %
Elastic Net S0	1.417	4.688	0.337	-0.399	-0.577
Elastic Net S1/S2	1.241	4.688	0.337	-0.399	-0.577
GBDT S0	3.511	4.420	1.558	-0.231	-2.275
GBDT S1	3.534	5.534	1.329	0.529	-1.199
GBDT S2	3.128	5.440	1.322	0.464	-1.199
Cash	1.316	4.808	0.186	N/A	0.000
SPY fixed shares	8.800	21.672	14.896	1.076	-19.261

Note: Validation and test portfolios are initialized independently. Returns and volatility are annualized; maximum drawdown is measured over the full test-period path. All strategies include applicable costs under the main setting.

GBDT S1 and S2 exceed cash by 0.726 and 0.632 percentage points in annualized return, respectively; GBDT S0 and all Elastic Net specifications underperform cash. Under common thresholds and costs, adding

cross-sector histories is associated with improved GBDT trading performance, while the linear model shows no incremental gain. Relationship summaries in S2 do not improve on S1.

GBDT S1 holds sector pairs in only 13.127% of test intervals and cash otherwise. This substantial cash allocation helps explain the strategy's low volatility and limited drawdown, alongside its modest return increment above cash.

Fixed-share SPY delivers substantially higher absolute returns alongside greater volatility and drawdown (Table 13), reflecting its continuous equity-market exposure.

7.4.2 Sensitivity to Cost and Financing Assumptions

Cost sensitivity covers eight scenarios: the main setting; execution fees of 0, 1 or 5 bp; annual borrowing fees of 0%, 2% or 3%; and an interest rebate on short-sale proceeds. Each alternative changes one factor while holding forecasts, thresholds and trading rules fixed. Independently reconstructed ledgers incorporate the effect of costs on subsequent position sizes.

GBDT S1's test net annualized return ranges from 5.342% to 5.822%, and S2's from 5.258% to 5.712%. Both remain above cash across all eight scenarios, while Elastic Net and GBDT S0 remain below it. The principal ranking is therefore unchanged within these settings.

7.5 Comparative Results for the SPY Overlay Strategy

7.5.1 Thresholds and Model Configurations

Table 14: Net annualized returns of the SPY overlay configurations

Model and specification	Validation 10/5 bp %	Validation 15/7.5 bp %	Test 10/5 bp %	Test 15/7.5 bp %
Elastic Net S0	7.888	9.137	21.728	21.884
Elastic Net S1	8.103	8.942	21.728	21.884
Elastic Net S2	8.103	8.942	21.728	21.884
GBDT S0	10.849	11.251	21.576	21.414
GBDT S1	11.434	11.083	21.851	22.723
GBDT S2	10.865	10.676	22.464	22.616
SPY with dividend reinvestment	9.047	9.047	22.019	22.019

Note: Each of the first six rows contains two trading configurations, giving 12 candidates. The final row is the common investment benchmark for both threshold pairs.

Threshold effects vary across models and periods. Lowering the entry threshold to 10 bp improves GBDT S1 and S2 during validation but reduces their test returns; GBDT S0 exhibits the reverse pattern. Elastic Net performs better at 15 bp in both periods. Greater participation therefore has no uniform net-return advantage.

Only three configurations exceed dividend-reinvestment SPY during testing: GBDT S1 and S2 at 15 bp, and GBDT S2 at 10 bp. GBDT S1 at 15/7.5 bp is the highest-return configuration among the 12 candidates and is examined below.

7.5.2 Risk and Relative Performance of the Selected Configuration

Table 15: Test-period performance of the selected overlay and SPY benchmarks

Metric	GBDT S1 15/7.5 bp	SPY with dividend reinvestment	SPY fixed shares
Net annualized return	22.723%	22.019%	21.672%
Cumulative net return	114.119%	109.586%	107.379%
Annualized volatility	15.576%	15.239%	14.896%
Maximum drawdown	-19.303%	-19.764%	-19.261%

The selected configuration earns a test net annualized return of 22.723%, exceeding the dividend-reinvestment benchmark's 22.019% by 0.704 percentage points. Terminal net asset value is 2.14119, compared with 2.09586 for the benchmark, implying 2.163% greater terminal wealth.

Annualized tracking error is 1.240% and the information ratio is 0.504. Volatility exceeds the benchmark by 0.337 percentage points, with a slightly smaller maximum drawdown (Table 15). The enhancement therefore

retains a risk profile close to continuous equity-market investment.

During validation, the same configuration earns a net annualized return of 11.083%, compared with 9.047% for SPY with dividend reinvestment. Figure 9 presents net asset value and relative wealth paths for both periods.

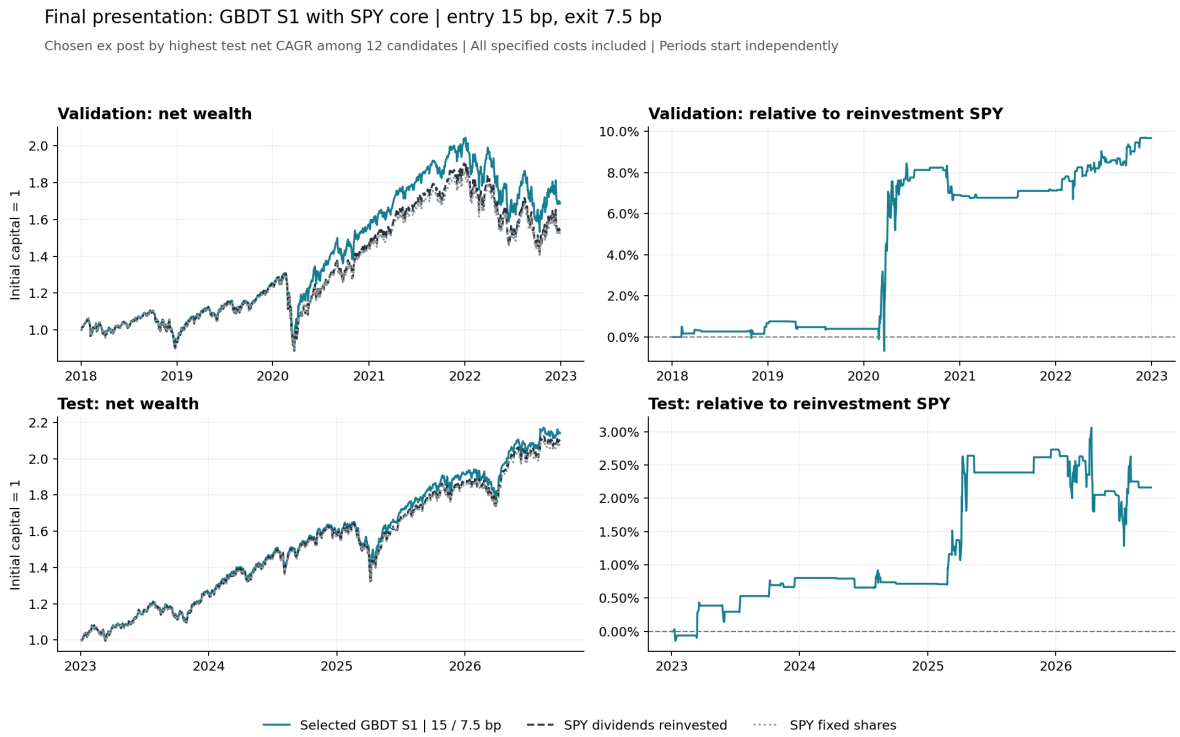


Figure 9: Selected overlay performance in validation and testing

The left panels show net asset value after costs, with initial capital normalized to 1. The right panels show strategy net asset value divided by that of SPY with dividend reinvestment, minus 1. Each period starts independently with initial capital of 1.

7.5.3 Position Activity and Implementation Costs

Sector participation is 13.127% of test intervals, comprising 74 holding episodes averaging 2.135 open-to-open intervals. Mean gross exposure is 102.321%, maximum post-trade gross exposure is 150%, and mean short exposure is 1.161%. The strategy consequently remains close to SPY for most intervals, with intermittent sector adjustments.

Annualized two-sided turnover is 9.547 times net asset value. Execution fees, stock borrowing fees and financing interest over the full test period amount to 1.691%, 0.068% and 0.354% of initial capital, respectively, totaling 2.113%. Execution fees are the largest explicit cost component.

7.6 Sources and Stability of Incremental Returns

7.6.1 Decomposing the Terminal Wealth Difference

The daily ledger decomposes terminal wealth differences into sector-pair contributions, capital-path effects and implementation costs. Sector contributions equal pre-cost profit or loss on actual sector holdings less that on SPY positions with matching signed dollar exposure. The capital-path term captures SPY profit or loss arising from different strategy and benchmark capital. Components are additive in units of initial capital, normalized to 1.

Note: Cost components are incremental expenses relative to SPY with dividend reinvestment. Dividing the terminal wealth difference by benchmark terminal net asset value gives the relative wealth increase of 2.163%.

Sector adjustments generate a positive pair contribution, while execution and holding costs absorb part of the increment. The capital-path component is also positive, reflecting the effect of earlier gains and losses on the amount of capital subsequently invested in the market.

At the sector level, XLP and XLU contribute approximately 3.450% and 2.340% of initial capital to test-period

Table 16: Test-period terminal wealth difference relative to SPY with dividend reinvestment

Component	Percentage of initial capital
Sector-pair contribution relative to SPY	4.7753
SPY contribution from capital-path differences	1.7751
Difference in cash interest	Approximately 0
Additional execution fees	-1.5956
Additional stock borrowing fees	-0.0680
Additional financing interest	-0.3538
Terminal wealth difference	4.5329

pair profit or loss. XLB and XLE contribute -1.303% and -1.061%, respectively.

7.6.2 Annual Variation and Holding-Episode Concentration

Table 17: Annual performance of the selected overlay and SPY with dividend reinvestment

Year	Strategy period return %	SPY reinvestment period return %	Relative wealth increase %
2023	27.216	26.205	0.801
2024	25.288	25.396	-0.087
2025	20.212	17.849	2.005
January–September 2026	11.752	12.376	-0.555

Note: Annual returns compound the actual intervals within each evaluation period. Relative wealth increase is $(1 + R_{\text{strategy}})/(1 + R_{\text{SPY}}) - 1$. The 2026 figures cover the period through 30 September.

Positive incremental performance is concentrated in 2023 and 2025. The strategy slightly underperforms the benchmark in 2024 and also trails it during the first nine months of 2026.

Of the 74 holding episodes, 41 have positive pair contributions. The largest positive episode accounts for 8.13% of total positive episode contributions, and the five largest together account for 29.79%.

Implementation: notebooks 05–06; artifact paths: Appendix B.

8 Conclusions

Cross-sector return histories provide limited, model-dependent predictive gains. Strong Elastic Net shrinkage produces identical test forecasts across S0–S2, with excess prediction losses concentrated in XLF. GBDT S1 has slightly better pooled forecast and trading point estimates than S0, but its test loss-improvement confidence intervals include zero and performance varies across years and sectors. The relationship summaries in S2 provide no consistent further improvement.

Economic performance depends on portfolio structure and implementation costs. GBDT S1's cash-based pairs earn a modest increment over cash, while the SPY overlay maintains market exposure and delivers a smaller increment relative to dividend-reinvestment SPY. Sector adjustments contribute positively to the reported terminal wealth difference, with execution fees the largest explicit cost. Revised data snapshots, changing sector classifications and opening-price execution assumptions also limit interpretation.

The overlay was introduced after test results had been examined, and its reported configuration was selected for the highest test return among twelve candidates. These results therefore constitute ex post exploration rather than independent evidence for a preselected strategy. Future work should evaluate fixed modeling and trading rules on subsequent data and examine whether signal strength remains sufficient after costs.

9 Team Contributions

YU ZIMU was responsible for the data acquisition, cleaning and feature construction code, the corresponding report sections, and the video narration explaining these parts.

ZHU YIXU was responsible for the Elastic Net training, forecast evaluation and diagnostic code, the corresponding report section, and its video narration. ZHU YIXU also consolidated the team's written contributions and prepared the final report's LaTeX layout.

CHEN DEDAN was responsible for the GBDT training, model selection and forecast evaluation code, the

corresponding report section, and its video narration.

YANG AIER was responsible for the portfolio construction, backtesting and performance analysis code, the corresponding report section, and its video narration.

YANG AIER and YU ZIMU were jointly responsible for recording the code rerun in a clean environment and for video editing.

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A Supplementary Elastic Net Comparisons

These are the validation-window and XLF comparisons discussed in the forecasting section. R^2 in these tables is expressed in percent. They are consolidated here to keep the main model comparison continuous.

Table 18: Validation R^2 across model families and training windows

Scheme	Family	Expanding	Rolling 5y	Rolling 3y
S0	Elastic Net	-0.0691%	-0.1508%	-0.4330%
S0	Ridge	+0.1133%	-0.0304%	-0.0953%
S1	Elastic Net	-0.0605%	-0.1881%	-0.5157%
S1	Ridge	+0.1288%	-0.0387%	-0.1911%
S2	Elastic Net	-0.0605%	-0.1900%	-0.5410%
S2	Ridge	+0.1307%	-0.0394%	-0.1731%

Table 19: XLF validation R^2 under different training windows

Scheme	Family	Expanding	Rolling 5y	Rolling 3y
S0	Elastic Net	-0.660%	-0.212%	-0.637%
S0	Ridge	+0.065%	+0.001%	-0.276%
S1	Elastic Net	-0.567%	-0.267%	-0.778%
S1	Ridge	-0.141%	-0.193%	-0.630%
S2	Elastic Net	-0.567%	-0.267%	-0.789%
S2	Ridge	-0.140%	-0.190%	-0.633%

B Reproducibility

B.1 Data and execution

Tables and figures match saved artifacts at commit `bc7b92f`, including archived portfolio runs under `results/backtest/`. The current fresh-run workflow writes separate `results/reproduction/` outputs. Report assembly did not retrain models or rerun portfolio simulations.

Elastic Net is fitted by cyclic coordinate descent (`selection='cyclic'`, tolerance 10^{-6}); GBDT uses histogram-based gradient boosting with squared-error loss. Both implementations set `random_state=5205`. The saved Elastic Net run records Python 3.12.10, NumPy 2.5.4, pandas 3.0.6 and scikit-learn 1.9.1 (`results/elastic_net/run_manifest.json`); these package versions are pinned in `requirements.txt`.

Use Python 3.12 and the repository's complete `requirements.txt`. From the repository root, a macOS/Linux setup is:

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m ipykernel install --user --name dsa5205 \
  --display-name "Python (DSA5205)"
python -m notebook
```

Select the matching kernel and run notebooks 01 through 06 in order: frozen-data verification, feature construction, Elastic Net, GBDT, pair backtesting and overlay evaluation. Fresh-run source hashes prevent mixing earlier outputs with changed inputs.

After forecasts have been regenerated, the portfolio stages can alternatively be invoked with:

```
python -B scripts/fresh_backtest_pipeline.py --stage all
python -B -m unittest discover -s scripts \
  -p test_fresh_backtest_pipeline.py -v
```

No machine-specific paths are needed. Preserve `data/raw/` and `notebooks/download_manifest.json` together. A fresh provider download can change historical data and is different from reproducing the frozen snapshots.

Detailed correspondence between report evidence and saved files is provided in the repository README under "Report artifact correspondence". Notebook and result paths use the repository root as their base.

B.2 Evaluation conventions

Forecast scoring uses 1,257 validation and 937 test holding intervals across nine sectors. Terminal forecasts without mature labels are not scored. Each portfolio phase starts independently. R^2 is a ratio unless marked in percent; annualized return is CAGR, while Sharpe and information ratios use daily arithmetic excess returns.